

# Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 1: Where Have the Fish Gone? Launching Our Investigation into Ecosystems

Teacher reference only. Do not project or hand out.

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| B  | A  | C  | D  | D  | C  | C  |

USE AFTER: PREDICT PHASE — Individual prediction and quick sketch showing living and non-living things in the river

**Q1. A learner sketches a river in Kenya to show what might be found in it. Which list has BOTH living and non-living things from a river?**

- A. Tilapia, frogs, birds, insects
- B. Tilapia, algae, water, sunlight**
- C. Water, oxygen, rocks, sand
- D. Algae, aquatic weeds, tilapia, Nile perch

**Correct answer: B.** Tilapia and algae are living, while water and sunlight are non-living. The other lists contain only living things or only non-living things.

USE AFTER: OBSERVE PHASE — Part A (Video): 3-column table 'What I see | What I hear | What I wonder'

**Q2. A learner watched a video of a Kenyan river and filled in the 'What I see' column. Which entry is a specific observation rather than a general statement?**

- A. Thick algae and aquatic weeds cover parts of the water**
- B. The river is in a bad state
- C. The environment has been spoiled
- D. People do not care about the river

**Correct answer: A.** Thick algae and weeds covering the water can be seen directly in the video. The other entries are opinions or general statements, not specific observations.

USE AFTER: OBSERVE PHASE — Part B (Reading): table matching each term to an example from the river in the video

**Q3. In a river, all the tilapia living in the river together are called a:**

- A. Community
- B. Habitat
- C. Population**
- D. Biome

**Correct answer: C.** A population is all the individuals of one species in an area, here all the tilapia in the river. A community includes many species together.

USE AFTER: OBSERVE PHASE — Part B (Reading): group table matching habitat and niche to river examples

**Q4. The river water and banks where the tilapia live is the tilapia's \_\_\_\_\_. The role the tilapia plays in the river, such as what it eats and how it lives, is its \_\_\_\_\_.**

- A. community; population
- B. niche; habitat
- C. population; community
- D. habitat; niche**

**Correct answer: D.** A habitat is the place where an organism lives. A niche is the role of the organism in its ecosystem. Learners often confuse the two.

USE AFTER: EXPLAIN PHASE — Claim-evidence-reasoning paragraph comparing first ideas ('fish are old; fish have no food') with the video and reading

**Q5. A group says, 'The fish are disappearing because they are old, or because they have no food.' They compare this idea with what was seen in the river video. Which is the best conclusion at this point?**

- A. The idea should be ignored because it was only a prediction
- B. The idea explains everything, so no more investigation is needed
- C. The video has already proved the cause, so the question is solved
- D. The idea explains only part of the evidence. The booming algae and weeds and the smell are still unexplained**

**Correct answer: D.** Old age or lack of food does not account for the thick algae and weeds or the smell. These observations remain unexplained and need further investigation.

USE AFTER: DQB PHASE — Individual sticky-note questions and sorting into categories on the Driving Question Board

**Q6. Learners are writing questions on sticky notes for the Driving Question Board about the river where fish are disappearing. Which question is the best one to investigate?**

- A. Who owns the river?
- B. Is the river dirty?
- C. How do algae and weeds affect the fish in the river?**
- D. Are the fish old?

**Correct answer: C.** A good Driving Question Board question is open and can be investigated. The first two can be answered yes or no, and ownership does not help explain the ecosystem.

USE AFTER: MODEL BUILDING PHASE — Exit Ticket (1): Define ecosystem in your own words

**Q7. Which sentence is the best definition of an ecosystem?**

- A. All the individuals of one species living in an area
- B. The place where one organism lives
- C. The living things in an area together with the non-living parts of their environment, interacting with each other**
- D. All the non-living things in an area, such as water, light and temperature

**Correct answer: C.** An ecosystem includes both living (biotic) and non-living (abiotic) components that interact in an area. The other choices describe a population, a habitat and abiotic factors only.